

Madhur Mehta

Heising-Simons QuRIOS Postdoctoral Fellow · University of California, Santa Barbara

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Research Profile

Theoretical physicist and scientific-software builder developing analytic and computational frameworks for complex physical systems. My research spans **black holes**, **gravitational waves**, **effective field theory**, **black-hole perturbation theory**, and **quantum gravity**, with an emphasis on connecting fundamental dynamics to testable observables. Alongside my research, I build software for scientific discovery and research workflows. I am now expanding this work toward **AI for scientific reasoning and discovery**, with particular interest in systems that can use mathematical, computational, and information-retrieval tools to solve difficult research problems.

Research Experience

University of California, Santa Barbara

Sep. 2025 – Present

Heising-Simons QuRIOS Postdoctoral Fellow

Santa Barbara, CA

- **Black-hole response and cavity effective field theory:** Co-developed a finite-radius effective framework for describing the response of compact gravitational systems and connecting internal dynamics to scattering and gravitational-wave observables.
- **Black-hole perturbation theory:** Developing gravitational response functions for perturbations of Schwarzschild black holes and extending the framework toward spinning black holes.
- **Dissipation and finite-frequency response:** Studying absorption, tidal interactions, fluxes, finite-frequency effects, and the relation between near-horizon dynamics and asymptotic observables.
- **Strong-field signatures of new physics:** Investigating how modifications of classical black-hole behavior can propagate into measurable scattering and waveform signatures.

The Ohio State University

Jan. 2021 – Jul. 2025

Ph.D. in Physics

Columbus, OH

Advisor: Prof. Samir D. Mathur

- **Universal black-hole thermodynamics:** Showed that sufficiently compact horizonless objects reproduce black-hole thermodynamic behavior under broad assumptions, identifying properties that do not depend explicitly on horizon formation.
- **Extremely compact objects:** Studied horizonless compact objects and extensions involving charge and other black-hole-like properties.
- **Electromagnetic entrapment in strong gravity:** Co-developed compact electromagnetic configurations trapped by strong gravitational fields that reproduce several macroscopic properties associated with black holes.
- **D1–D5 conformal field theory:** Studied lifting of superconformal descendants and multi-mode states relevant to the microscopic description of black-hole states.

- **Black-hole information problem:** Investigated microscopic and semiclassical approaches to the information problem, including fuzzball physics, quantum extremal surfaces, and related questions in gravitational entropy.

Indian Institute of Science

Sep. 2019 – Mar. 2020

Master's Thesis Research

Bengaluru, India

Advisor: Dr. Chethan Krishnan

- Studied quantum extremal surfaces, entanglement wedge reconstruction, generalized entropy, and the Page curve in AdS/CFT approaches to the black-hole information problem.

Johannes Gutenberg University

May – Jul. 2018

DAAD WISE Research Fellow

Mainz, Germany

Advisor: Prof. Sebastian Boeser

- Characterized the timing response of wavelength-shifting optical modules using picosecond light sources and silicon photomultiplier detectors.

University of Tokyo

Jun. – Aug. 2017

UTRIP Research Fellow

Tokyo, Japan

Advisor: Prof. Shuji Hasegawa

- Studied the growth and characterization of $(\text{Bi}_x\text{Sb}_{1-x})_2\text{Te}_3$ topological-insulator materials and associated experimental instrumentation and data analysis.

IISER Bhopal

Apr. – May 2017

Summer Research Intern

Bhopal, India

Advisor: Prof. Surajit Saha

- Studied solid-state spectroscopy techniques, including Raman spectroscopy, for condensed-matter systems.

Scientific Software

Naksh Lens [\[Find it on the App Store\]](#)

2026 – Present

- Designed and built an iOS research-discovery application that retrieves recent arXiv papers and ranks them according to a researcher's active themes, followed authors, publication history, and relevance feedback.
- Developed personalized ranking, research-lens explanations, author prioritization, multi-category arXiv scanning, local caching, and research-journal workflows.
- Integrated arXiv and INSPIRE-HEP data and developed research-focused discovery workflows designed to help researchers identify papers relevant to their current interests.
- Improved feed reliability and update detection by separating retrieval logic for new/cross-listed papers

from replacement papers, then validating results against arXiv daily listings and resolving missed-update edge cases.

- Redesigned author scanning and batching to substantially reduce retrieval time for larger followed-author lists while preserving robustness across categories and legacy arXiv identifiers.

Education

The Ohio State University

2021 – 2025

Ph.D. in Physics

Columbus, OH

Advisor: Prof. Samir D. Mathur

Thesis: *Through the Horizon and Back: Lifting the Veil on Black Hole Thermodynamics*

The Ohio State University

2022

M.S. in Physics

Columbus, OH

Indian Institute of Science Education and Research Bhopal

2015 – 2020

Integrated B.S.–M.S. in Physics

Bhopal, India

Recent Publications

Full publication record: [INSPIRE-HEP](#)

1. S. B. Giddings and **M. Mehta**, *Deriving Effective Descriptions and Signal Predictions for Dynamical Gravitational Systems*, arXiv:2606.19435, 2026.
2. S. Chakraborty, **M. Mehta**, and G. Patashuri, *Extremal Curves in Single-Trace $T\bar{T}$ -Holography*, *Journal of High Energy Physics* 11 (2025) 128.
3. **M. Mehta**, *Extreme Compactness, Extreme Gravity: Higher-Derivative Corrections to ECOs*, arXiv:2505.09049, 2025. *Honorable Mention, Gravity Research Foundation Essay Competition, 2025.*
4. S. D. Mathur and **M. Mehta**, *The Fuzzball Paradigm*, arXiv:2412.09495, 2024. *Invited chapter for The Black Hole Information Paradox.*
5. S. D. Mathur and **M. Mehta**, *The Universal Thermodynamic Properties of Extremely Compact Objects*, *Classical and Quantum Gravity* 41 (2024) 23, 235001.
6. P. Heidmann and **M. Mehta**, *Electromagnetic Entrapment in Gravity*, *Journal of High Energy Physics* 03 (2024) 046.
7. M. R. R. Hughes, S. D. Mathur, and **M. Mehta**, *Lifting of Superconformal Descendants in the $D1$ – $D5$ CFT*, *Journal of High Energy Physics* 04 (2024) 129.
8. S. D. Mathur and **M. Mehta**, *The Universality of Black Hole Thermodynamics*, *International Journal of Modern Physics D* 32 (2023) 14, 2341003. *Third Prize, Gravity Research Foundation Essay Competition, 2023.*

Selected Talks and Presentations

- *From Interior Dynamics to Exterior Observables: Cavity EFT and Finite-Radius Response Functions*, QuRIOS Collaboration Meeting, July 2026.
- *Extremely Compact Objects and Their Potential Observational Signatures*, High Energy Physics Lunch Seminar, Kavli Institute for Theoretical Physics, October 2025.
- *Universality of Black-Hole Thermodynamics*, QuRIOS Collaboration Meeting, September 2025.
- *What Are Fuzzballs and Extremely Compact Objects?*, TASI Summer School, University of Colorado Boulder, June 2025.
- *Universal Thermodynamics of Extremely Compact Objects*, Strings 2024, poster presentation, June 2024.
- *The Ohio State University David DeMartini Poster Competition*, poster presentation, May 2024.
- *Universal Thermodynamics of Extremely Compact Objects*, APS Eastern Great Lakes Section Meeting, invited speaker, April 2024.
- *The Ohio State University Hayes Forum*, invited talk, March 2024.

Teaching and Mentoring

Physics Summer Research Program Advisor

May – Jul. 2024

The Ohio State University

- Mentored an undergraduate research student and introduced them to general relativity and theoretical-physics research.

Graduate Teaching Assistant

2021 – 2022

The Ohio State University

- Physics 1251: Calculus-Based Electricity and Magnetism, Waves, Optics, and Modern Physics.
- Physics 1201: Algebra-Based Electricity and Magnetism, Waves, Optics, and Modern Physics.
- Physics 5300: Theoretical Mechanics.

Scientific Communication, Outreach, and Service

Stargazing Guide, Alta Vista Tours [\[View Stargazing Tour\]](#)

2026 – Present

Lead small-group guided stargazing experiences in the Santa Barbara area, using telescopic observations to introduce guests to planets, stars, star clusters, nebulae, galaxies, constellations, and connections between astronomy, mythology, and the human experience of the night sky.

Time Before Space [YouTube Channel]

2021 – Present

Founder and host of a science-communication YouTube channel on fundamental physics and foundational questions. Interviews include Juan Maldacena, Abhay Ashtekar, Gary Horowitz, Niayesh Afshordi, Sera Markoff, and Samir Mathur.

Other Outreach and Service

- Public talk, “Perspectives of Gravity,” Upper Arlington Library, July 2023.
- Physics outreach volunteer, Ohio State Fair Youth Exploration Space, July–August 2023.
- Journal article reviewer, *International Journal of Modern Physics D*, 2021.
- Judge, Metro Capstone Symposium, Metro Early College High School, 2021.

Honors and Awards

- Heising–Simons QuRIOS Postdoctoral Fellowship, 2025–present.
- Honorable Mention, Gravity Research Foundation Essay Competition, 2025.
- The Ohio State University Presidential Fellowship, 2024–2025.
- First Prize, The Ohio State University David DeMartini Poster Competition, 2024.
- Third Prize, Gravity Research Foundation Essay Competition, 2023.
- DAAD WISE Fellowship, 2018.
- University of Tokyo Research Internship Program Fellowship, 2017.
- Asian Science Camp Representative for IISER Bhopal, 2016.
- Bharat Ratna Prof. C. N. R. Rao Award, 2015.
- INSPIRE-SHE Scholarship, Department of Science and Technology, India, 2015–2020.

Technical Skills

Physics: General relativity; black-hole perturbation theory; effective field theory; gravitational-wave physics; QFT in curved spacetime; black-hole thermodynamics; compact objects; conformal field theory.

Scientific Computing: Mathematica; Python; analytic and perturbative calculations; symbolic computation; numerical prototyping.

Programming: Python; Java; C; HTML/CSS; prior experience across coursework and independent projects.

Software and Product Development: AI-assisted software development; mobile application prototyping; API integration; iterative testing and debugging; deployment and release workflows.

Research Software and Data Tools: scientific-information retrieval; personalized ranking; relevance evaluation; scientific-data integration; retrieval validation; research-workflow prototyping.